Exercises week 3

8CM10 - Introduction to Quantum Physics and Structure of Matter

Schrödinger Equation

1. Consider a (one-dimensional) particle that moves in a constant potential energy $U(x) = c$ for all x.
    a. Solve the Schrödinger equation for this case. Show that there are only bounded solutions $\Psi(x)$ if the total energy E satisfies $E \geq c$.
    b. What is the wavelength of $\Psi(x)$ for a given value of E, with $E \geq c$?
    c. What is the de Broglie wavelength of a particle with a kinetic energy $K = E - c$ for a given value of E, with $E \geq c$? Compare this value to the value you found for (b).
    d. What is the physical difference between this particle and a free particle?

2. Let $\Psi(x)$ be a solution of the one-dimensional Schrödinger equation for a given potential $U(x)$ with energy E. Show that the function $\widehat{\Psi}(x) = A\Psi(x)$ is also a solution of the one dimensional Schrödinger equation with potential $U(x)$, with the same E, for any values of the constant A. What does this show about normalizing a wave function?

3.
    a. Explicitly show by differentiation that $\psi(x) = Ae^{kx} + Be^{-kx}$ solve the Schrödinger equation for a region where $V = V_0 < \infty$ (a constant) and $E < V_0$. How is the variable k defined?
    b. Suppose we would take the $\lim_{V_0 \to \infty} V_0$, what does this imply for k?

4.
    a. Explicitly show by differentiation that both $\psi_a(x) = Acos(kx) + Bsin(kx)$ and $\psi_b(x) = Ce^{ikx} + De^{-ikx}$ solve the Schrodinger equation for a region where $V = V0$ (a constant) and $E > V0$. How is the variable k defined?
    b. Use Eulers identity for complex exponentials to show that $\psi_a(x)$ and $\psi_b(x)$ are equivalent solutions by relating A, B, C and D. Please remember that A, B, C and D are in general complex coefficients.

5. For the one dimensional infinite square well with boundaries set at $x = 0$ and $x = a$ the wave function corresponding to the n'th stationary state is given by the following function:

$$\psi(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), & x \in [0, a] \\ 0 \quad , & x < 0 \quad \text{or} \quad x > a \end{cases}$$

   a. Compute the probability density for all the allowed states.
   b. For $n = 1, 2, 3$ plot the probability density for the particle. Afterwards give interpretations of the plots! (If you find this difficult without numerical values, you may used $a = 5.29 \cdot 10^{-11}\text{m}$ (= Bohr radius to 3 significant digits)
   c. Compute the expected value of position for a particle in eigenstate n. Give your physical interpretation of this result.
      Remember: $\langle x \rangle = \int \psi^*(x)\, x\, \psi(x)\, dx$
      Hint for calculation: Substitute $y = \frac{n\pi x}{a}$ *and use* $\sin^2(y) = \frac{1-\cos(2y)}{2}$. *Afterwards use partial integration to reduce the integral to an integral of an elementary function.*

6. Compute in detail the wave functions and corresponding energies for a two-dimensional rectangular box with size L_x by L_y. Give the expression and calculate the normalization constant. Use this calculation to determine the wave function for a square box with side length L.

7. A particle is represented by the following wave function:

$$\psi(x) = \begin{cases} 0, & x < -\frac{L}{2} \\ C\left(\frac{2x}{L} + 1\right), & -\frac{L}{2} < x < 0 \\ C\left(\frac{2x}{L} + 1\right), & 0 < x < \frac{L}{2} \\ 0, & x > \frac{L}{2} \end{cases}$$

   a. Use the normalization condition to find C.
   b. Evaluate the probability to find the particle in an interval of width $0.010L$ at $x = L/4$ (that is, between $x = 0.245L$ and $x = 0.255L$. (No integral is necessary for this calculation.)
   c. Evaluate the probability to find the particle between $x = 0$ and $x = +\frac{L}{4}$.
   d. Find the average value of x and the rms-value of x: $x_{rms} = \sqrt{(x^2)_{av}}$

8. In this exercise we consider a particle with mass m in a potential $U(x)$ given by

$$U(x) = \begin{cases} \infty & \text{if} \quad x < 0 \\ 0 & \text{if} \quad 0 \leq x \leq L \\ U_0 & \text{if} \quad x > L \end{cases}$$

where U_0 is some positive number, see also Figure 1. Hence at the left wall of the well (at $x = 0$) the potential goes from 0 to ∞ and at the right wall (at $x = L$) the potential goes from 0 to U_0.

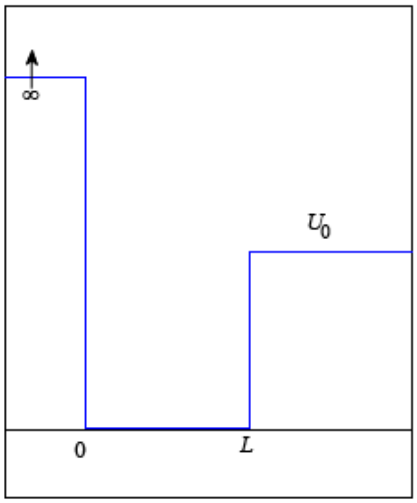


Figure 1: The potential $U(x)$.

In this exercise we try to find solutions to the Schrödinger equation for this situation with a total energy E with $0 < E < U_0$.

a. Show that in classical ("Newton") mechanics a particle with total (=potential + kinetic) energy $E < U_0$ cannot escape from the well.
b. What is the solution $\Psi(x)$ of the Schrödinger equation for $x < 0$?
c. What is the general solution $\Psi(x)$ of the Schrödinger equation for $0 \leq x \leq L$?
d. Show that continuity of $\Psi(x)$ in x = 0 implies that the solution for $0 \leq x \leq L$ must be of the form $\Psi(x) = A \sin kx$.
e. Give k^2 in terms of the energy E.
f. What is the general solution $\Psi(x)$ of the Schrödinger equation for $x > L$?
g. Show that the behavior for $x \to \infty$ requires that the solution for $x > L$ is of the form $\Psi(x) = Ce^{-ax}$ with $a > 0$.
h. Give a^2 in terms of the energy E and the potential value U_0.
i. The continuity of $\Psi(x)$ and of $\frac{d\Psi(x)}{dx}$ in $x = L$ leads to two conditions. Give these conditions.
j. Show that part (i) implies that k and a must be related by

$$a = -k \frac{\cos kL}{\sin kL} \qquad (1)$$

k. Show that parts (e) and (h) imply that a and k also must satisfy the condition

$$a = \sqrt{\widehat{U_0} - k\text{^}2} \qquad (2)$$

And give $\widehat{U_0}$.

A solution $\Psi(x)$ of the given form exists if there are a and k values that satisfy the two equations (1) and (2). To find out whether solutions exist, we first introduce new (dimensionless) variables x and y by $x = kL$ and $y = aL$.

l. Show that in terms of the new unknowns and y the equations are

$$y = -x\frac{\cos x}{\sin x}$$

$$y = \sqrt{F^2 - x^2}$$

With $F^2 = \frac{2mU_0L^2}{\hbar^2}$.

m. The graph of the function $y = -x\frac{\cos x}{\sin x}$ is the red line in Figure 2. To find the intersections with the graph $y = \sqrt{F^2 - x^2}$ recall that the last graph is the supper part of a circle with radius F. How many solutions are there for $F = 3$? And for $F = 6$? *Hint: use a compass*

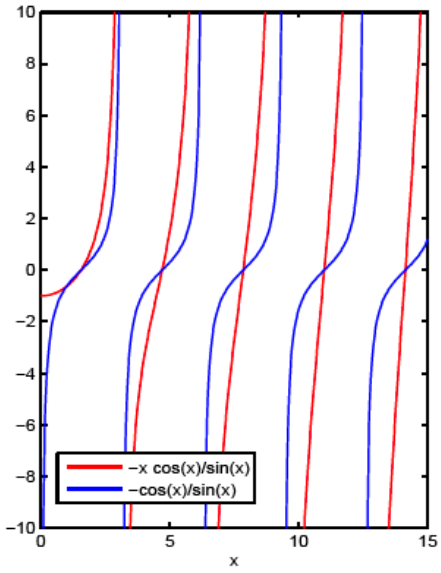


Figure 2: The graphs of $y = -\cos(x)/\sin(x)$ (blue) and $y = -x\cos(x)/\sin(x)$ (red).

n. Suppose the particle is an electron, $L = 1.0 \cdot 10^{-10}\text{m}$ and $U_0 = 380$ eV. How many discrete energy levels are there is this case?